

Ryan Cho

Senior Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full stack software engineer with 7+ years building REST APIs, React and TypeScript clients, and SQL-backed financial systems at scale. Led end-to-end delivery of transaction evaluation, risk scoring, and third-party integration workflows across SaaS platforms serving millions of users. Expertise in schema design, query optimization, A/B experimentation, and mentoring engineers on API design and modern JavaScript standards.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and owned **Java-based REST services** powering income verification and transaction evaluation flows, reducing average API latency by **35%** through query tuning and schema refinements across PostgreSQL-backed stores.
- Built **React and TypeScript** web clients for new lending products, delivering end-to-end feature ownership from API contracts through interaction design and **A/B-tested** conversion improvements.
- Integrated **third-party financial APIs** (credit, payroll, banking) into unified onboarding pipelines serving **2M+ monthly active users** with idempotent retry logic and structured observability.
- Designed **relational database migrations** and analytical SQL models supporting risk scoring and predictive eligibility checks used by underwriting and operations teams.
- Implemented **automated visual regression testing** in CI pipelines, catching UI defects pre-release and lowering customer-reported layout issues by **40%**.
- Partnered with product and customer success to ship **internal automation tools** that deflected **25%** of repeat support tickets through self-serve status and exception handling workflows.
- Led **GraphQL gateway** evaluation and pilot for read-heavy dashboards, simplifying client data fetching while maintaining REST compatibility for legacy mobile clients.
- Mentored **three junior engineers** on API design, TypeScript patterns, and production debugging, elevating team velocity and on-call readiness during platform expansions.
- Deployed containerized services via **Docker and Kubernetes**, standardizing release workflows and improving deployment frequency for revenue-critical user journeys.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Java REST microservices** for trade settlement and reference data APIs consumed by front-office and risk applications across global markets infrastructure.
- Optimized **SQL stored procedures and indexes** on Oracle schemas, improving batch reconciliation runtime by **28%** for daily regulatory reporting workflows.
- Built **React dashboards** for operations teams to monitor transaction exceptions, translating business rules into reusable UI components with TypeScript strict mode adoption.
- Implemented **predictive validation algorithms** for monetary transaction screening, reducing false-positive alerts while maintaining compliance with internal risk thresholds.
- Integrated external **market data and verification APIs**, adding circuit breakers and structured logging to harden services against third-party outages.
- Authored **database migration scripts** and data lineage documentation supporting audit-ready schema changes across multiple downstream analytics consumers.
- Collaborated with cross-functional stakeholders in agile squads to prioritize API enhancements that improved internal trader workflow completion times by **15%**.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Contributed to **React Native** mobile features for restaurant inventory tracking, implementing ES6+ patterns and component tests aligned with design system guidelines.
- Assisted in building **Node.js REST endpoints** for device sync and offline-first data reconciliation used by field teams across pilot deployments.
- Wrote **SQL queries and lightweight migrations** on PostgreSQL to support reporting on check-in volumes and venue utilization metrics.
- Participated in **A/B experiments** on onboarding copy and layout, instrumenting events and summarizing results for product review sessions.
- Automated **visual snapshot tests** for critical checkout screens, integrating them into pull request checks to reduce manual QA cycles.
- Documented API contracts and setup guides that accelerated **two subsequent intern cohorts** ramping on the shared codebase.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Delivered **responsive web pages** for nonprofit program listings using JavaScript, HTML, and CSS, improving mobile readability for community stakeholders.
- Connected front-end forms to **REST backend services** for volunteer registration, validating payloads and surfacing actionable error states.
- Maintained **MySQL-backed content modules**, writing ad hoc analytical queries to summarize engagement by campaign and region.
- Implemented accessible navigation and **motion-enhanced UI transitions** per design feedback, increasing time-on-page across key landing sections.
- Coordinated with remote stakeholders to iterate releases under tight deadlines, establishing reusable templates for future site sections.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, React Native, TypeScript, JavaScript (ES6+), HTML5, CSS3, Redux, Webpack, Interaction Design, Visual Regression Testing

Backend: Java, Spring Boot, REST APIs, GraphQL, Node.js, Microservices, Third-Party API Integration, Transaction Processing

Databases: PostgreSQL, Oracle, MySQL, SQL Schema Design, Migrations, Query Optimization, Analytical SQL

Cloud & DevOps: Docker, Kubernetes, AWS, CI/CD, Container Orchestration, Infrastructure as Code

Monitoring & Tools: Datadog, Grafana, Splunk, Jest, Cypress, Git, Jenkins

Workflow & PM: Agile/Scrum, A/B Testing, Technical Mentorship, Cross-Functional Collaboration, Jira, Confluence